

The total number of marks available for this examination paper is 80. It contributes 20% to the Advanced GCE in Physics.

Calculator

It is recommended that students have access to a scientific calculator for this paper.

Formulae sheet

Students will be provided with the formulae sheet shown in *Appendix 8: Formulae*. Any other physics formulae that are required will be stated in the question paper.

5.3 Thermal energy

This topic covers specific heat capacity, internal energy and the ideal gas equation.

This topic may be taught using applications that relate to, for example, space technology.

Students will be assessed on their ability to:	Suggested experiments
109 investigate, recognise and use the expression $\Delta E = mc\Delta\theta$	Measure specific heat capacity of a solid and a liquid using, for example, temperature sensor and data logger
110 explain the concept of internal energy as the random distribution of potential and kinetic energy amongst molecules	
111 explain the concept of absolute zero and how the average kinetic energy of molecules is related to the absolute temperature	
112 recognise and use the expression $\frac{1}{2} m\langle c^2 \rangle = \frac{3}{2} kT$	
113 use the expression $pV = NkT$ as the equation of state for an ideal gas	Use temperature and pressure sensors to investigate relationship between p and T Experimental investigation of relationship between p and V